



Shortest Path, Homework 12 - Solution

1 Shortest-Paths Properties.

1. Let G be a directed graph, $\delta(u, v)$ the length of a shortest path from u to v , and $d[v]$ an estimator for $\delta(s, v)$ for a fixed source $s \in V$. Assume that $d[v]$'s are initialized as follows:

$$d[v] = \begin{cases} \infty & v \neq s \\ 0 & v = s \end{cases}$$

And that $d[v]$ values are updated by relaxations only. Prove on your own the **Upper-bound property** stated as follows: After the application of any sequence of relaxation steps, we have $d[v] \geq \delta(s, v)$.

Solution.

1. By induction on the i th relaxation step, we present only the induction step. If $d[v]$ was changed after the step, then it was set to be $d[u] + w(u, v)$ for some $u \in V$ such that $(u, v) \in E$. However:

$$\begin{aligned} d[u] + w(u, v) &\geq \delta(s, u) + w(u, v) \\ &\geq \delta(s, v) \end{aligned}$$

Where the first inequality uses the induction assumption for the $i - 1$ previous relaxation steps, and the second follows from the triangle inequality.

2 Bellman-Ford

1. Let G be a directed, connected graph that may contains negative cycles. Prove that if G is indeed contains such then Bellman-Ford detects it. An implementation of Bellman-Ford is given in Algorithm 1.

Solution.

1. Since any vertex is reachable from the source s , after $|V| - 1$ iterations, a relaxation sequence over the edges on the path from s to any $v \in V$ is applied, and the value of $d[v]$ is set to be a real number, i.e., not ∞ .

Algorithm 1: Bellman–Ford.

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1 for  $v \in V$  do
2    $d[v] \leftarrow \infty$ 
3    $\pi[v] \leftarrow \text{null}$ 
4 end
5  $d[s] \leftarrow 0$ 
6 for  $i \in [n - 1]$  do
7   for  $(u, v) \in E$  do
8     if  $d[v] > d[u] + w(u, v)$  then
9        $d[v] \leftarrow d[u] + w(u, v)$ 
10       $\pi[v] \leftarrow u$ 
11     end
12   end
13 end
14 for  $(u, v) \in E$  do
15   if  $d[v] > d[u] + w(u, v)$  then
16     return Negative Cycle
17   end
18 end

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Suppose that $v_1, v_2, \dots, v_k = v_1$ is a negative cycle. This means that:

$$\sum_{i=2}^k w(v_{i-1}, v_i) < 0$$

Assume that the algorithm doesn't detect the cycle, meaning the condition at Line 15 is not satisfied for any of the edges. Namely for all $v \in V$ and $(u, v) \in E$ we have $d[v] < d[u] + w(u, v)$. Note that we use the fact that all $d[v]$ s are real numbers, so the comparison is well-defined. Namely, at the final check, we don't compare infinities. Then:

$$\begin{aligned}
\sum_{i=2}^k d[v_i] &< \sum_{i=2}^k d[v_{i-1}] + \sum_{i=2}^k w(v_{i-1}, v_i) \\
&= \sum_{i=2}^k d[v_i] + \sum_{i=2}^k w(v_{i-1}, v_i) \\
&\Rightarrow 0 < \sum_{i=2}^k w(v_{i-1}, v_i)
\end{aligned}$$

Where, for the first pass, we used the fact that $v_2, \dots, v_k$ is a cyclic permutation of $v_1, \dots, v_{k-1}$.

On the other hand, if there is no negative cycle, then any path that has more than $|V| - 1$ edges contains a positive cycle, and therefore has a length equal to or greater than the path obtained from it by removing the cycle. Since lines (6)-(12) have already applied the relaxation steps of a shortest path in

order, the additional relaxation at Line 15 can't improve the estimator, and therefore the condition will not be satisfied.